OLYMPIC MEDAL ANALYSIS DASHBOARD

Project Report

Project: Olympic Medal Analysis using Tableau

Project link: Open the Tableau dashboard

This report describes the purpose, design, development, and evaluation of an interactive dashboard for exploring Olympic medal information. The dashboard link above is the project output.

1. INTRODUCTION

1.1 Project Overview

The Olympic Medal Analysis project uses Tableau to present Olympic medal information through an interactive dashboard. It is designed to help users explore medal distribution and compare results across available categories, such as countries, sports, events, and Olympic editions, depending on the fields contained in the dataset. Visual summaries make the information easier to navigate than a raw table.

1.2 Purpose

The purpose of the project is to organize Olympic medal data into clear visualizations, support comparisons, and make patterns easier to identify. Interactive views allow users to explore the information and focus on categories relevant to their questions.

2. IDEATION PHASE

2.1 Problem Statement

Olympic medal records can be extensive and difficult to compare when viewed as rows of data. Users may find it challenging to understand how medals are distributed among countries or how results differ across sports and editions. The project addresses this by presenting the data in an organized Tableau dashboard.

2.2 User Needs

Users need a simple way to explore medal records, compare categories, and interpret the displayed information. Clear labels, consistent visual encoding, and interactive controls help reduce the effort required to navigate the dataset.

2.3 Brainstorming

The dashboard concept focuses on using charts and filters to turn medal records into an accessible visual summary. Tableau was selected because it supports interactive charts and dashboard layouts.

3. REQUIREMENT ANALYSIS

3.1 User Journey

A user opens the dashboard, reviews the overview, selects or filters a category of interest, compares the displayed medal information, and uses the resulting view to explore the dataset.

3.2 Solution Requirements

- A structured Olympic medal dataset in a format Tableau can read.
- Tableau Public for data exploration and visualization.
- Clear chart titles, labels, and legends.

- An interactive dashboard that allows users to explore the available categories.

3.3 Data Flow

Olympic medal data is loaded into Tableau, reviewed and organized, then used to build individual visualizations. The visualizations are assembled into a dashboard and published for interactive viewing.

3.4 Technology Stack

- Visualization tool: Tableau Public
- Data source: Olympic medal dataset
- Output: Interactive Olympic Medal Analysis Dashboard
- Platform: Tableau Public

4. PROJECT DESIGN

4.1 Problem–Solution Fit

The dashboard addresses the difficulty of interpreting large medal tables by presenting the information visually and enabling users to compare categories interactively.

4.2 Proposed Solution

The proposed solution is a Tableau dashboard that brings together relevant medal visualizations in one place. Users can explore the available medal records through the dashboard's charts and controls.

4.3 Solution Architecture

Dataset → data preparation and field selection → Tableau worksheets → dashboard assembly → published interactive dashboard.

5. PROJECT PLANNING & SCHEDULING

The work can be organized into the following stages: identify the dataset and project questions; inspect and prepare the data; create individual worksheets; assemble and format the dashboard; test the visualizations and interactions; publish the dashboard; and prepare the report.

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Functional Testing

- Check that each visualization loads and displays the intended fields.
- Check that dashboard filters and interactive selections respond as expected.
- Confirm that titles, labels, legends, and tooltips are understandable.
- Verify that the published dashboard opens from the project link.

6.2 Performance and Usability

Review the dashboard for readable labels, consistent formatting, useful comparisons, and reasonable loading behavior. Testing should be repeated after any changes to the data or dashboard.

7. RESULTS

7.1 Output

The project output is an interactive Tableau dashboard for exploring Olympic medal information. The published dashboard is available here: Olympic Medal Analysis Dashboard.

7.2 Interpretation

The dashboard supports visual exploration and comparison of the medal data. Specific numerical findings should be reported only after checking the values and filters in the published dashboard.

8. ADVANTAGES AND LIMITATIONS

Advantages

- Makes medal information easier to explore visually.
- Supports comparisons across available categories.
- Combines multiple views in an interactive dashboard.
- Can be accessed through a published Tableau link.

Limitations

- The analysis depends on the completeness and quality of the source dataset.
- The dashboard can only answer questions represented by the available fields.
- Visual summaries do not, by themselves, explain the causes of medal outcomes.

9. CONCLUSION

The Olympic Medal Analysis project demonstrates how Tableau can transform structured medal records into an interactive visual dashboard. It provides a convenient way to explore and compare the available Olympic medal information.

10. FUTURE SCOPE

- Add more filters and comparison views where supported by the dataset.
- Include additional Olympic editions or data fields when reliable data is available.
- Improve tooltips and explanatory annotations.
- Refresh the dataset and dashboard as new records become available.

11. APPENDIX

Project Link

https://public.tableau.com/app/profile/mahima.patidar6985/viz/OlympicMedalAnalysisDashboard_17935318863380/OympicMedalAnalysisDashboard

GitHub Repository

https://github.com/mahimapatidar2487-cmd/Global-Olympic-Sports-Performance-Analysis